

FICTIONAL PETROCHEM TRAINING PLANT

Training Example – SOP & RCA

Document ID	SOP-PS-618-REV3 (Training Example)
Title	Emergency Isolation Procedure for T-412 Distillation Overhead Accumulator
Facility	Utilities & Distillation Training Unit
Effective Date	October 10, 2024

Purpose

This fictional document is provided only for engineering training and formatting practice. It does not describe a real facility or operational procedure.

Equipment Data

Parameter	Value
Primary Asset	T-412 Distillation Overhead Accumulator
Associated Pump	P-412C Reflux Pump
Normal Pressure	18.5 bar
MAWP	25.0 bar
Operating Temperature	165 C
High Pressure Alarm	22.8 bar
PSV	PSV-412A (23.8 bar)
Emergency Isolation Valve	EIV-412

Operating Procedure

- Verify permit and instrumentation.
- Purge with nitrogen.
- Start auxiliary systems.
- Start reflux pump gradually.
- Monitor pressure, temperature and level.
- Reduce feed and isolate on high-pressure alarm.

LOTO

- Stop pump.
- Isolate power.
- Close inlet and outlet valves.
- Apply locks/tags.
- Drain and vent before maintenance.

Illustrative RCA

Incident ID: TRN-2024-017

Scenario: Simulated pressure increase after temporary reflux pump trip.

Illustrative Root Cause: Reduced condenser performance due to exchanger fouling.

Illustrative CAPA: Clean exchanger, improve predictive monitoring, shorten inspection intervals, and retrain operators.

Notice

This document is fictional and intended solely for training.